

Homework 4 Answers

EPI 525

Book exercises

3.20 Area under the curve, Part II

! Important

For this problem:

- Make a sketch of the normal distribution curve with the mean and 1 sd away from the mean clearly labeled, and the area representing probability of interest shaded in.
- Calculate probabilities using R (You can hand write the code you used to find your answer if you do not want submit typed R code or a Quarto document.)

a

87.08%

b

57.14%

c

$6.22^{-14}\%$

d

38.29%

3.22 Triathlon times

a

13.8%

b

37.74%

c

3354.05 seconds

d

6295.21 seconds

3.28 Arsenic poisoning

0.26 g/dl and 6.14 g/dl

3.32 Chickenpox, Part III

a

Expected value: 108, sd: 3.29

b

0.223

1.28 Mix-and-match

Describe the distribution in the histograms below and match them to the box plots.

- a) 2
- b) 3
- c) 1

R exercises

R exercise 1

Below you will be using a dataset from Gapminder to complete a few R exercises.

Load all the packages you need below here

You don't need to do it all at once, you can add more libraries as you realize you need them.

Import dataset

Import the dataset called "Gapminder_2011_LifeExp_CO2.csv" You can find it in the student files under Data then Homework. You will need to download the file onto your computer, and use the correct file path to import the data.

Make a histogram

Using `ggplot2`, make a histogram of the variable `CO2emissions`.

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 4 rows containing non-finite outside the scale range  
(`stat_bin()`).
```

